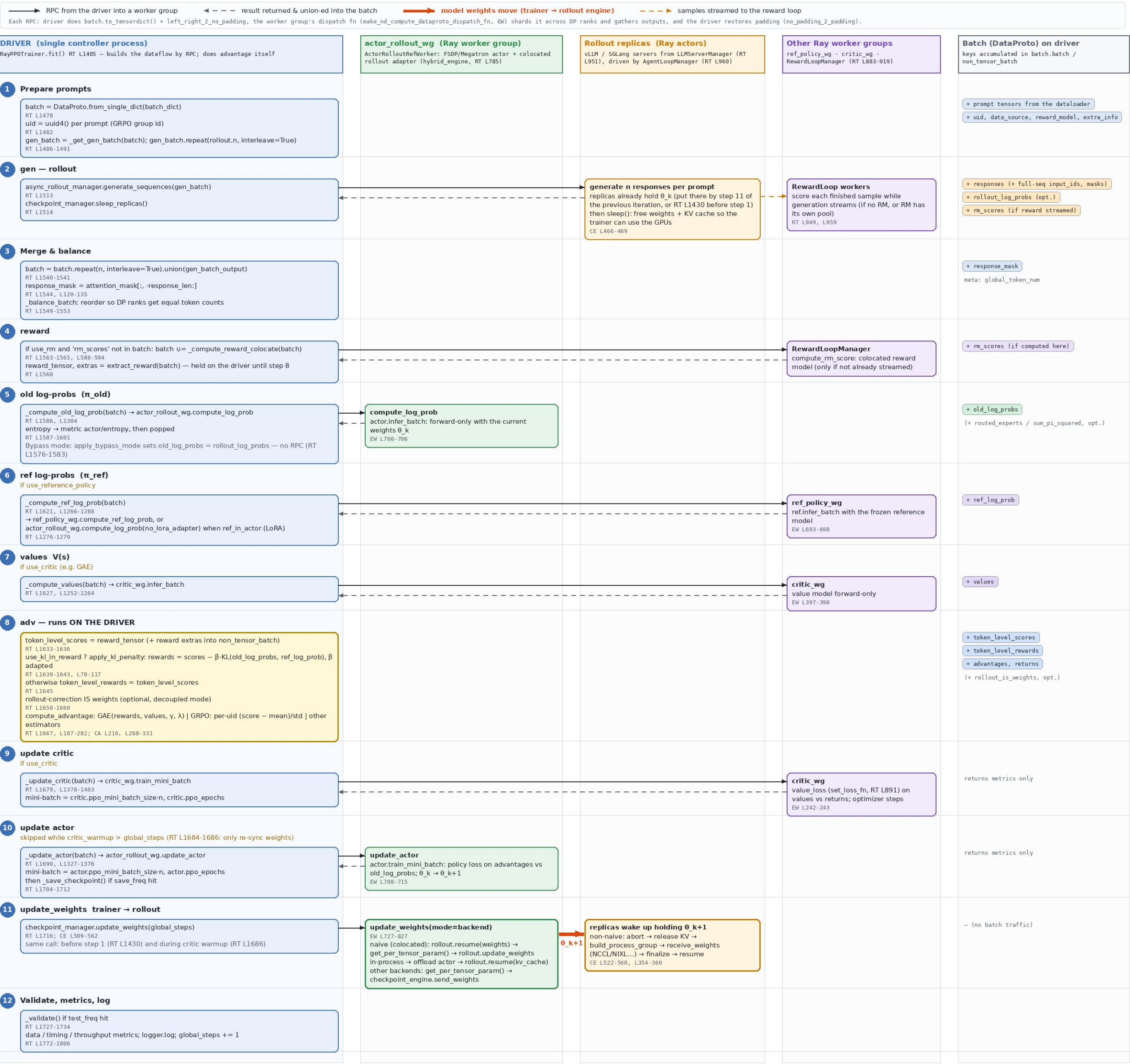


verl — what one PPO / GRPO training step does (RayPPOTrainer.fit, single-controller)

verl @ fbb4b3a8bf63 (2026-09-30). RT = verl/trainer/ppo/ray_trainer.py EW = verl/workers/engine_workers.py CE = verl/checkpoint_engine/base.py CA = verl/trainer/ppo/core_algos.py
Note: RayPPOTrainer is marked @deprecated at this commit (RT L285: legacy trainer, use trainer.use_v1=True); this is still the loop in the pinned file.



actor_rollout_wg and the rollout replicas share the same resource pool / GPUs (hybrid engine, RT L785-793, L951-953): that is why rollout sleeps after gen (step 2) and only wakes when weights are pushed (step 11).

Reading the flow

- The driver runs steps 1, 3, 8 and 12 itself (data prep, merge/balance, KL-in-reward and advantage). Every other step is a blocking RPC into a Ray worker group, so steps run strictly in order 2 → 11.
- GRPO (critic off): steps 7 and 9 are skipped and the advantage is group-normalized per uid. PPO with GAE: the critic gives values (7) and is trained on returns (9) before the actor (10).
- Weights move once per step, at 11: from the actor's training engine (FSDP/Megatron) to the rollout engine (vLLM/SGLang). Everything else moves the batch (DataProto ↯ TensorDict).